

Sinoda Ghebrehiwet Tesfay

Software Engineer · Backend & ML Infrastructure · 6 Years Experience

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📍 Vancouver, Canada



Software Engineer with 6 years across backend, ML infrastructure, and full-stack development. Core strengths in **Python**, **Rust**, and **Go**, async API design, distributed systems, and real-time ML serving. Seeking a Backend or Platform Engineering role.

PROFESSIONAL EXPERIENCE

Esklate : Backend Engineer, Afro-Chat

2023 – 2026

- Directed the migration from Flask and MongoDB to an async **FastAPI + SQLAlchemy** architecture, reducing API latency by 35%.
- Built real-time ML serving streaming live training telemetry to browsers at **20 fps over FastAPI WebSockets**, with a SQLite experiment registry and on-demand model endpoints.
- Shipped the recommendation engine for **Bazar**, an AI-driven marketplace, using TensorFlow and scikit-learn on an automated Python + Selenium pipeline.

Kuraz Tech Engineering PLC : React Frontend Developer

2021 – 2023

- Cut application load times by **40%** through code optimization and lazy loading within a 30-person frontend team.
- Drove the migration to **Next.js** with SSR and SSG, improving SEO and lifting user engagement by 25%.
- Mentored junior developers via code review and pair programming, reducing onboarding time by 30%.

Esklate : Full Stack Developer, Internal Tooling

2020 – 2021

- Built a Slack-integrated automation platform for Google Sheets workflows, adopted by **200+ employees** across four time zones.
- Designed a timezone-aware notification system on Redis and Celery job queues, reducing missed deadlines by 30%.
- Orchestrated highly available Go microservices on **Kubernetes**, cutting deployment times by 60% at 99.9% uptime.

PROJECTS

RustV OS Lab Bare-Metal RISC-V Operating System

A bare-metal Rust (no-std) kernel booting on QEMU without BIOS or SBI: trap handling, Sv39 virtual memory, preemptive scheduling, user-mode syscalls, a from-scratch virtio-MMIO DMA block driver, and a persistent filesystem.

Rust RISC-V Assembly QEMU Sv39 VM Virtio-MMIO

GAN Observatory Real-Time GAN Training Observatory

Streams training telemetry to an interactive browser dashboard at ~20 fps via FastAPI and WebSockets, with a dependency-light NumPy GAN built from scratch, live hyperparameter updates, and automatic mode-collapse detection.

Python NumPy TensorFlow/Keras FastAPI WebSockets Docker

ImmuStore DB Distributed Database Engine Built from Scratch

A distributed database engine in pure Python: copy-on-write B+ tree storage with $O(\log n)$ reads, MVCC with shadow-paging commits and crash recovery, a Redis RESP async server, and Raft-based replication with leader election and failover.

Python asyncio B+ Trees MVCC Raft Redis RESP

AI-RAD Healthcare Screening & Records Platform

Hospital COVID-19 screening and digital records with role-based access for doctors, nurses, and admins. Won **1st place** in the 2020 UNDP & MINT COVID-19 Innovation Challenge, securing a 580,000 ETB grant.

React Node.js Firebase Firestore RBAC

SKILLS

Languages	Python, Rust, Go, Java, TypeScript, JavaScript, Dart, SQL, RISC-V Assembly
Backend	FastAPI, Node.js, SQLAlchemy, REST, WebSockets, asyncio, Celery, Redis, MongoDB, SQLite
ML & Systems	TensorFlow/Keras, scikit-learn, NumPy, distributed systems, Raft, MVCC, concurrency
Frontend	React, Next.js, Flutter, Tailwind CSS, Unity
DevOps	Docker, Kubernetes, GitHub Actions, CI/CD, pytest, JUnit 5, Git

EDUCATION

Software Engineering

Addis Ababa Institute of Technology
2014 – 2019 · Addis Ababa, Ethiopia

Coding Academy (DS & Algorithms)

A2SV
2022 – 2023 · Addis Ababa, Ethiopia